🚙 Automobile Data Analysis Report

(Based on Master Table: p_master_data_analysis_200)

📊 1. Objective

The objective of this project is to analyze automobile data by integrating multiple datasets into a master table and deriving meaningful insights using data visualization techniques in Power BI. The analysis focuses on pricing, sales, fuel trends, insurance status, service costs, and regional distribution.

🗂 2. Dataset Overview

The master dataset (`p_master_data_analysis_200`) was created using SQL by combining:

- Car Data
- Insurance Data
- Sales Data
- Owner Details
- Service History

This unified dataset enables comprehensive analysis across multiple business dimensions.

📊 3. KPI Dashboard Summary

Key performance indicators provide a quick overview of the dataset:

- **Total Cars:** 200

Average Price: ₹26.79 Lakh
Average Sale Price: ₹27.37 Lakh
Average Service Cost: ₹11,121

🎯 Insight

The average sale price is slightly higher than the listed price, indicating a **strong resale market** and good value retention of vehicles.

🚙 4. Brand Analysis

📊 Visualization

Bar Chart showing count of cars by brand.

🔍 Findings

Brands like **Mahindra and Maruti** have the highest number of vehicles.
A few brands dominate the dataset.

🧠 Insight

The automobile market shows a **concentrated structure**, where a limited number of brands capture the majority of market share.

⛽ 5. Fuel Type Distribution

📊 Visualization

Bar Chart / Pie Chart showing count of cars by fuel type.

🔍 Findings

Electric and CNG vehicles have the highest counts.
Diesel vehicles have the lowest presence.

🧠 Insight

There is a clear shift toward **eco-friendly and cost-efficient fuel options**, reflecting changing consumer preferences and environmental awareness.

🛡️ 6. Insurance Status Analysis

📊 Visualization

Donut Chart showing Active vs Expired insurance.

🔍 Findings

Nearly equal distribution between **Active and Expired** insurance policies.

🧠 Insight

This highlights a **significant business opportunity** for insurance providers to target expired policies for renewal campaigns.

💰 7. Price vs Sale Price Comparison

📊 Visualization

Clustered Column Chart comparing average price and average sale price by brand.

🔍 Findings

Sale prices are generally **higher than purchase prices**.

🧠 Insight

This indicates a **strong resale market**, where vehicles maintain or increase their value over time.

📅 8. Sales Trend Analysis

📊 Visualization

Line Chart showing sales over time.

🔍 Findings

Sales fluctuate across different periods.

🧠 Insight

Identifying peak sales periods helps businesses optimize:

Inventory planning
Marketing campaigns
Pricing strategies

🔧 9. Service Analysis

📊 Visualization 1

Bar Chart showing average service cost by service type.

🔍 Findings

Certain services (e.g., Full Service, Battery Replacement) have higher costs.

📊 Visualization 2

Column Chart showing total service cost by service center.

🔍 Findings

Some service centers generate significantly higher revenue.

🧠 Insight

Service-related data helps in:

Identifying high-revenue services
Improving pricing strategies
Enhancing operational efficiency

🌍 10. Location Analysis

📊 Visualization

Map / Bar Chart showing number of cars by city.

🔍 Findings

Higher concentration of car ownership in urban areas.

🧠 Insight

Demand for automobiles is **higher in metropolitan cities**, indicating stronger purchasing power and urban dependency on vehicles.

🎛️ 11. Interactive Dashboard Features

The dashboard includes slicers for:

Brand
Fuel Type
City
Year

🎯 Benefit

These filters allow users to perform **dynamic and customized analysis**, enhancing usability and decision-making.

🧠 12. Overall Business Insights

Market Concentration
Few brands dominate the automobile market.
Sustainability Trend
Increasing adoption of Electric and CNG vehicles.
Insurance Opportunity
Large number of expired policies indicates untapped revenue potential.
Strong Resale Market
Vehicles maintain or increase value over time.
Service Revenue Potential
High-cost services contribute significantly to revenue.
Urban Demand
Higher vehicle ownership in cities.

📊 13. Conclusion

This analysis provides a comprehensive view of the automobile dataset by combining multiple data sources into a unified master table. The dashboard highlights key trends in brand dominance, fuel preferences, pricing, insurance, and service operations.

The insights derived can help businesses make informed decisions related to:

- Inventory management
- Pricing strategies
- Customer targeting
- Service optimization

📌 Final Presentation Statement

"This dashboard provides a complete view of the automobile dataset. It highlights brand dominance, fuel trends, insurance risks, and service costs. The visuals clearly show a shift toward eco-friendly vehicles and strong resale value, along with a significant opportunity in insurance renewals."